Patrick Iben

Curriculum vitae

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Research profile

Applied mathematician and biostatistician with seven years of clinical biostatistics in contract research and twelve years in research. Works to eliminate needless ambiguity, risk, and suffering in decisions that rest on data, by stating in advance how a claim could fail and reporting exactly what was measured and what was not. Develops statistical frameworks and inspectable software across clinical biometrics, AI evaluation, wearable sensing and public-data monitoring. Independent work connects model validation and reproducibility with scientific software, interactive demonstrations and educational companions. Open to full-time and consulting roles in research science, research engineering, AI evaluation, biostatistical methods and decision-support systems.

Professional experience

Biostatistician II | Syneos Health | Remote | December 2021 to October 2026

- Lead biostatistician on early-phase clinical pharmacology studies from December 2022, across first-in-human, SAD/MAD, crossover, bioequivalence and bioavailability designs.
- Owned statistical protocol sections, generated randomization schedules and provided statistical input to clinical study reports.
- Prepared statistical analysis plans and mock tables, listings and figures; created or reviewed specifications for analysis datasets and outputs using CDISC SDTM/ADaM.
- Performed PK/PD analyses and reviewed tables, figures and listings before delivery.
- Reviewed study data during trials and reviewed annotated CRFs and database designs to support planned analyses.

Associate Biostatistician | Emmes | Rockville, MD | September 2019 to December 2021

- Lead and support statistician on federally sponsored Phase 1 and Phase 3 clinical trials focused on opioid use disorder.
- Wrote protocol statistical sections, statistical analysis plans, DSMB reports and final statistical reports.
- Built dynamic progress reporting and SAS/R tools for data-management and analysis problems.
- Harmonized multi-clinic EHR data and provided statistical support to external research partners.

Research Assistant and Teaching Assistant | University of Colorado Anschutz Medical Campus | Aurora, CO | August 2016 to July 2019

- Harmonized multi-state hospital databases containing millions of records and hundreds of derived variables to estimate pediatric nursing-care costs; designed the datasets for reuse in later studies.
- Consulted on study design and sample size for doctoral nursing candidates.
- Taught graduate biostatistics review sessions and office hours.

Earlier research appointments | 2014 to 2016

- **Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers.** Analyses for the Gates Foundation-funded Multi-State Collaborative assessment of student learning outcomes.
- **Data Analysis Fellow, Yale School of Child Development.** Evaluated assistive technologies for students with print disabilities.
- **Senior Data Analyst, SCSU Office of Assessment and Planning.** Developed predictive models of student success and retention.

Education

- **MS, Biostatistics**, Data Science Analytics minor, University of Colorado Anschutz Medical Campus, 2019. GPA 3.62.

- **BS, Mathematics**, Applications of Mathematics, Southern Connecticut State University, 2016. Magna cum laude; GPA 3.78.

Publications and manuscripts

Submission and review statuses as of 26 October 2026.

Contamination-resilient cluster randomized trial design under spatial interference: a simulation study of contamination control. *Statistics in Medicine*, submitted.

- Developed a design framework for cluster randomized trials in which treatment effects spill between nearby sites, and tested it by simulation on 57 real malaria survey sites in Uganda.
- Preprint doi:10.5281/zenodo.23486176.

What do within-site comparisons of AlphaGenome Atlas predictions measure? An audit at ClinVar sites in 11 genes. *Human Genomics*, submitted.

- Audited what an AI genomics model's variant comparisons actually measure, separating benchmark accuracy from clinical meaning, with replication across genes.
- Preprint doi:10.5281/zenodo.23543154.

A false floor: Refuting our own information-limit claim in a lethal identification task. Submitted to PLOS ONE, 22 October 2026.

- Overturned our own earlier claim of a hard information limit in a high-stakes identification task, using stronger baselines and checks for overlap with the models' training data.
- Preprint doi:10.5281/zenodo.23519299.

Anticipating network contagions: when directedness outperforms connectedness. *Applied Network Science*, with editor.

- Showed that targeting the units that transmit a contagion can beat targeting the hardest-hit ones, using a measure of directional asymmetry across seven contagion datasets (interventions simulated).

Which data modality recognizes a gene set? A domain-agnostic consilience engine. *PeerJ*, in peer review.

- Built an open-source tool that compares how well different kinds of biological data recognize the same gene groups, with statistical testing and replication on nervous-system and immune gene maps.

No emotion-specific image-computable code in the human left amygdala. Peer Community In, preprint review.

- Tested whether the human amygdala carries an emotion-specific visual code across three brain-imaging datasets and found no evidence for one, with conclusions limited to the samples and models tested.
- Preprint doi:10.5281/zenodo.21422754.

Published work

- **Acknowledged contributor:** developed data extraction and normalization programs for the study dataset in Osterhage KP, Hser YI, Mooney LJ, Sherman S, Saxon AJ, Ledgerwood M, Holtzer CC, Gehring MA, Clingan SE, Curtis ME, Baldwin LM. Identifying patients with opioid use disorder using International Classification of Diseases (ICD) codes: challenges and opportunities. *Addiction*. 2024;119(1):160-168. doi:10.1111/add.16338.
- **Co-author:** Ben-Avie M, Rosenwald L, Depastino R, Iben P. How can we help young adults with autism thrive in the workplace? *The Conversation*, 2016.

Teaching and research communication

- Authored six concise study guides and six complete educational companions tied to exact manuscript versions. All twelve editions are publicly available. The teaching sets include worked examples, answered exercises, glossaries and source-coverage maps.
- Connected manuscripts to code, data deposits, interactive demonstrations and teaching editions through a public research index. The CRT complete companion spans 97 pages; the other studies have their own full methods walkthroughs.
- Developed interactive statistical demonstrations and domain-specific explanations of the ten parameters used in response planning, supporting hover, keyboard focus and touch interaction.

Research engineering and scientific software

Trial biostatistics toolkit and eCRF review apps

A public toolkit for biostatistics consultants: runnable SAS and R templates, dashboards and QC checks for the work a biostatistician does for a client across an early-phase trial, from the protocol's statistical sections to CSR text. It runs from static files or local R, so client data never leave the analyst's machine; the example studies are synthetic.

- Built a complete worked example of a bioequivalence study: one synthetic trial taken from sample size and randomization through the analysis plan, analysis datasets, pharmacokinetic analysis, tables and report text, all regenerated by a single script.
- Built safety-monitoring tools for first-in-human studies: liver and cardiac screening rules, a Safety Review Committee packet and an interactive dashboard that runs in the browser.
- Built a quality-control harness that recomputes key results a second way and is tested against deliberately planted errors.
- Wrote SAS and R program libraries for standard tables, listings and figures across common early-phase designs, plus simulation studies of two regulatory checks.
- Built four browser-based apps for reviewing clinical trial data (medical coding, de-identification, participant risk and central monitoring) and published their measured limitations.

Public Data Consoles

- Built 32 self-contained, browser-only decision-support consoles spanning hazards, infrastructure, health-system strain, public health, access to care and markets; each labels every input as measured, modeled or demonstration. Examples include river-flood propagation, earthquake consequences, U.S. grid strain, NHS Scotland hospital strain, CDC surveillance and the World Bank global-health atlas.
- Integrated public observations, source-issued forecasts, published estimates and dated snapshots; implemented scheduled data publication, schema and timestamp checks, fallback behavior and source-specific freshness displays. Visitors need no provider API key.
- Adapted the ten-parameter planning framework across all 32 consoles. Response plans prioritize supported actions using the selected data, with review timing, escalation triggers, public/operator tracks and explanations of each parameter's practical relevance.
- Added mobile usability and keyboard/touch-accessible explanations; used catalog, unit and browser checks to verify source handling, plans and unavailable-feed states.

Coastal Water Resilience

- Unified my two research methods in Coastal Water Resilience: the ten-metric diagnostic suite from my contamination-resilient cluster randomized trial design, combined with directed source/sink topology accounting from my directed-contagion research, applied to live NOAA, NHC and NWS coastal forecasts across 131 published station locations, with terrain, evacuation zones and Census context driving projected impacts and response plans.
- Joined official public wastewater-station inventories to evacuation zones, USGS terrain and source-defined sewer basins, alongside NOAA/NWS observations and forecasts. Linked projected-impact screens to the same station scope used by the ten-parameter response plans.
- Incorporated Census population context and published annual treatment flows while distinguishing geographic exposure from operational outages, customer losses and hydraulic connectivity. The synthetic exercise remains a separately labeled demonstration.
- Connected parameters to phase-specific actions, explaining why a diagnostic matters at a particular timepoint and when to recheck or escalate.

Autonomic Telemetry

- Built a personal arousal-biofeedback system for work and rest: a multivariate arousal estimate combining heart rate, heart-rate variability (RMSSD, SDNN), EEG beta/alpha and covariance distance against a two-minute personal reference, with prediction-error event detection; a local Mac-to-iPhone bridge streams live input from a Muse S headband and a Hume Band, and a Signal Lab replays recordings exactly.
- Implemented source identity, freshness, signal-quality and channel-coverage controls, plus missing-channel handling and replayable feature exports. The physiological-pattern estimate is experimental.

- Built a separate Signal Lab with numerical lenses, analytic checks and regression tests. Live wearable input runs through the local bridge.

Resilience Telemetry

- Applied my directed-contagion research to cross-asset market volatility: spillover networks, transmitter stability and historical replay, tested against held-out baselines that track the loudest and the most persistent assets.
- Integrated public market and macroeconomic data with source timestamps and interpretation limits. The ridge-regularized VAR/FEVD implementation is an exploratory adaptation of the network-contagion research; the manuscript's non-negative VAR intervention experiments are documented separately.

Consilience Engine

- Packaged gene-set analysis as an installable Python library and configurable CLI with offline cache support. Constructed modality-specific graphs from InterPro, Human Protein Atlas, STRING and Gene Ontology sources, with permutation-null separation and class-recovery analyses.
- Applied the framework to a 410-gene nervous-system map and a 132-gene immune replication. Added unit, property and metamorphic tests, clustering-sensitivity analyses, numerical tolerance checks and reproducibility manifests.

Programming and evaluation skills

- **Programming and engineering:** R (tidyverse, pharmaverse and Shiny), Quarto, SAS (Base, Macro, STAT, SQL), Python, SQL, JavaScript, LaTeX, Git, Docker, high-performance computing, public API integration and browser testing.
- **AI evaluation:** numeric grounding, provenance auditing, retrieval evaluation, model comparison, synthetic-oracle checks and reproducibility testing.

Clinical expertise

- **Therapeutic areas:** cardiovascular, dermatology, immunology, neurology (MS, ALS, Parkinson's, Alzheimer's), oncology (hematologic malignancies and solid tumors), psychiatry (opioid use disorder and schizophrenia), and respiratory.
- **Study designs:** first-in-human, SAD/MAD, crossover, fixed sequence, parallel group, adaptive, placebo-controlled, double-blind, open-label and cohort.
- **Statistical methods:** PK/PD and NCA, bioequivalence, survival and longitudinal analysis, missing-data imputation, simulation, permutation and resampling inference, multiplicity, cluster randomized designs, causal inference under interference (spillover and contamination, directed network interventions), consilience testing across independent data modalities, calibration and decision curves.
- **Clinical standards and tools:** CDISC SDTM/ADaM, MedDRA and WHODrug coding, Veeva Vault, TLF specifications and review.